

Small Language Models Relevant to ChatEast

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Model	Params	Core Langs	Key Architecture	Notable Traits
InkubaLM-0.4 B	422 M	5 Bantu + EN/FR	8L-2048H-32-GQA	Llama-family variant; vocab 61 k; 8-bit OK
MobileLLM-125 M	125 M	EN baseline (re-tunable)	Deep-thin, weight share, GOA, tied embeddings	≤1 GB RAM inference; CPU-friendly
MobiLlama-0.5 B	500 M	EN baseline	22Lx1024 or 8Lx2048 param share	Llama-derived light decoder
SmolLM-135 M	135 M	EN (multi-lang fine-tune)	Standard Llama (light stack)	GGUF quant ready (<0.5 GB)
Swahili-RoBERTa-base-v2	110 M	Swahili	Encoder-only RoBERTa	Fast embeddings / NER
AfriBERTa-small	125 M	11 African langs	Encoder-only	Multilingual sentence vecs

Key Take-aways for ChatEast:

- SLMs cut GPU/CPU cost: InkubaLM-0.4 B runs at 8-bit in ~6 GB VRAM; MobileLLM-125 M on Raspberry-Pi-class edge.
 - African-centric vocab (Inkuba) reduces token budget by $\approx 20\%$ for Swahili, Luganda, Hausa.
 - Grouped-Query Attention + weight-sharing (MobileLLM) halves KV-cache $\rightarrow$ fits SMS micro-instance.
 - Quant-ready GGUF (SmolLM, Inkuba) enables offline kits for schools.
 - These models already outperform larger generic LLMs on AfriXNLI / AfriMMLU benchmarks.
- Deployment mix: edge (MobileLLM-125 M) for quick SMS, core GPU (InkubaLM-0.4 B) for deep RAG answers; SmolLM-135 M as re-ranker.